$a = -\omega^2 x$, $10 = (2\pi/T)^2 (0.050) \rightarrow T = 0.44$ s

15 **A** At displacement = $\frac{A}{2}$, $u^2 = \omega^2\left[A^2 - \left(\frac{A^2}{4}\right)\right]$

At displacement = 0, $v^2 = \omega^2 A^2$

Therefore $v = \frac{2}{\sqrt{3}}\ u$

16 **A** From graph, resonant freq. $f_0 = 1.5$ Hz, amplitude $x_0 = 16.4$ cm = 0.164 m.